

Technology Stack

Date	15 July 2026
Team / Submission ID	
Project Name	Emotion Detection & Learning Support Engine
Submitted By	Venu Gopal (Sole Team Member / Team Lead)
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Streamlit	Enables a fast, Python-native interactive UI (forms, tabs, charts) without writing separate HTML/JS, ideal for an ML-heavy solo project.
2	Backend / Application Logic	Python 3, modularised under src/ (preprocessing, prediction, logging, utils)	Keeps the emotion-detection pipeline reusable and testable, and matches the language of the ML libraries used.
3	Sequence Model	TensorFlow / Keras BiLSTM (models/bilstm/bilstm.keras)	Lightweight recurrent model that captures word-order context for emotion classification with fast CPU inference.
4	Transformer Model	PyTorch + HuggingFace Transformers, fine-tuned BERT (models/bert_emotion_model_final)	Higher-capacity contextual model used as a second opinion, letting the app compare and cross-check predictions.
5	NLP Preprocessing	NLTK (stopwords, tokenisation)	Standard, well-supported library for cleaning and tokenising free-text learner input before feeding the BiLSTM.
6	Generative AI	Google Gemini 2.5 Flash (google-generativeai SDK)	Fast, low-latency LLM used to turn the detected emotion into a natural-language, field-specific learning response.
7	Data Handling / Visualization	pandas, Plotly Express	pandas manages the interaction log and dataframes; Plotly renders the pie, line and bar charts in the analytics dashboard.
8	Configuration & Secrets	python-dotenv (.env file)	Keeps the Gemini API key out of source control while still making it available to the app at runtime.
9	Storage	Local CSV files (emotion_response_examples.csv) and pickled tokenizers / label encoders	Simple, dependency-free persistence appropriate for a single-user prototype; avoids the overhead of a full database.
10	Version Control	Git / GitHub	Tracks source history and is the submission mechanism for this project (public repository link).